Bias and MSE
WMS §8.1 - 8.2, SJ §4.3

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Introduction

Estimator : function
Estimate : 얻어진 값.

- In many cases, populations are charaterized by numerical descriptive measure called parameters.
- Ω: parameter space = all possible values of θ
- θ: unknown parameter in a model
- $Y_1, \cdots, Y_n$: data random sample from population
  $\hat{\theta} = d(Y_1, \cdots, Y_n)$: estimator of θ
- Many different estimators may be obtained for the same parameter.
- How can we establish criteria of goodness to compare statistical estimators?

⇒ $\boxed{\text{MSE}}$

ex) population mean - center of distribution
· sample mean sample variance
· sample median. ------

Bias

- Point estimation is similar to firing a gunshot at a target.
- Drawing a single sample and compute an estimate for a parameter $\Leftrightarrow$ firing a single shot at the target.
- We cannot evaluate the "goodness" of shooter based on only one shot. We must observe the results many times under the same setting. $\Rightarrow$ We look at the frequency distribution of the values of estimates in repeated sampling.

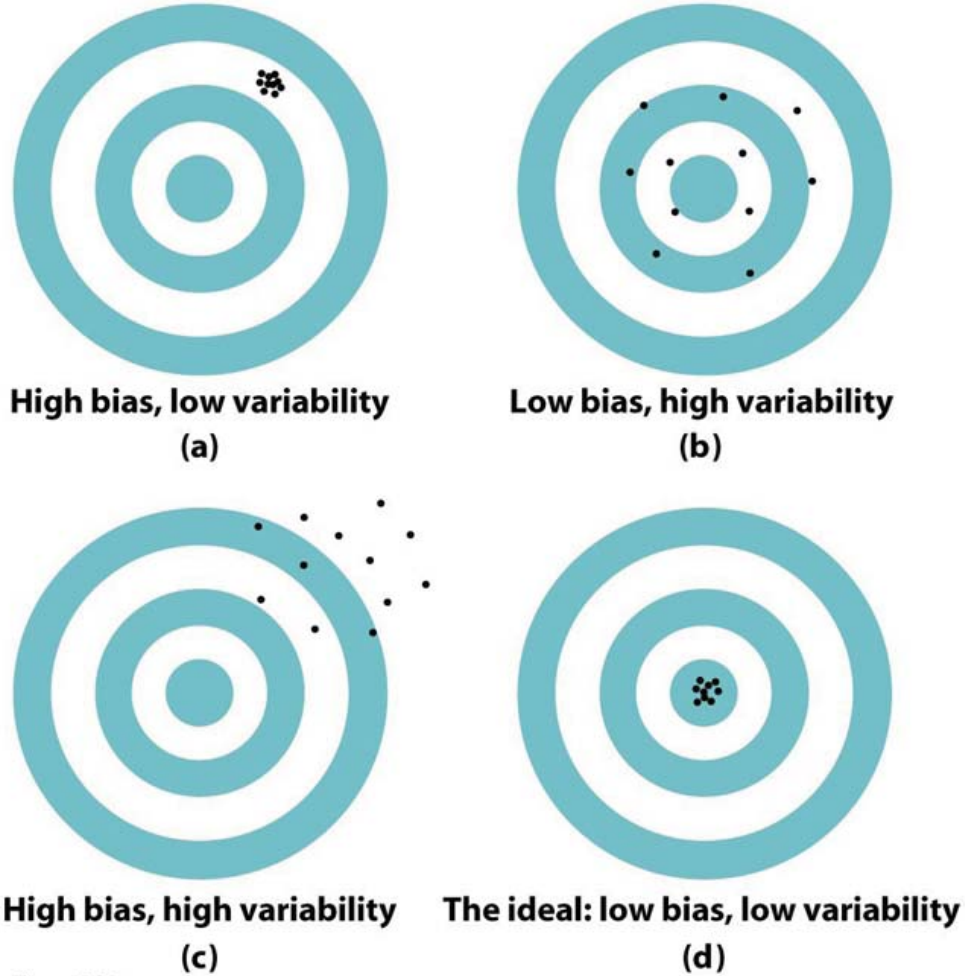

Figure 3-12
Introduction to the Practice of Statistics, Fifth Edition
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Bias

θ: target parameter, $\hat{\theta}$: estimator

Definition (Bias)

Bias of $\hat{\theta}$ as an estimator of θ is

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta.$$

$\hat{\theta}$ is an unbiased estimator of θ if $E_\theta(\hat{\theta}) = \theta$ for all $\theta \in \Omega$, otherwise, biased.

Ex. $Y_1, \cdots, Y_n$ iid with mean μ and variance σ^2

$$\Rightarrow E(\bar{Y}) = \mu, E(S^2) = E(\frac{1}{n-1}\sum(Y_i - \bar{Y})^2) = \sigma^2.$$

$\star$ $S = \overset{\dagger}{\sqrt{S^2}}$ is not UE of σ, in general.

$Y_1 \cdots Y_n$: iid (μ, σ^2)

- $E(\bar{Y}) = E(\frac{1}{n}\sum Y_i) = E(\frac{1}{n}Y_1 + \cdots + \frac{1}{n}Y_n)$

$$= E(\tfrac{1}{n}Y_1) + \cdots + E(\tfrac{1}{n}Y_n)$$

$$= \frac{1}{n}\underbrace{E(Y_1)}_{\mu} + \cdots + \frac{1}{n}\underbrace{E(Y_n)}_{\mu} = \mu$$

$$B(\bar{Y}) = E(\bar{Y}) - \mu = 0$$

- $E(S^2) = E\left(\frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right)$

$$\sum (Y_i - \bar{Y})^2 = \sum (Y_i^2 - 2\bar{Y}Y_i + (\bar{Y})^2)$$

$$= \sum Y_i^2 - 2\bar{Y}\underbrace{\sum Y_i}_{= n\bar{Y}} + n(\bar{Y})^2$$

$$= \sum Y_i^2 - n(\bar{Y})^2$$

$$= E\left(\frac{1}{n-1}\sum Y_i^2 - \frac{n}{n-1}(\bar{Y})^2\right)$$

$$= \frac{1}{n-1}\sum \underline{E(Y_i^2)} - \frac{n}{n-1}E((\bar{Y})^2)$$

$$= \frac{n}{n-1}E(Y^2) - \frac{n}{n-1}E((\bar{Y})^2)$$

$$E(Y^2) = Var(Y) + [E(Y)]^2$$

$$= \sigma^2 + \mu^2$$

$$Var\left(\frac{1}{n}Y_1 + \cdots + \frac{1}{n}Y_n\right)$$

$$= Var\left(\frac{1}{n}Y_1\right) + \cdots + Var\left(\frac{1}{n}Y_n\right)$$

$$= n \cdot Var\left(\frac{1}{n}Y_1\right) = n \times \frac{1}{n^2} Var(Y)$$

independence

$$= \frac{1}{n} \times \sigma^2$$

$$E\left((\bar{Y})^2\right) = Var(\bar{Y}) + \left[E(\bar{Y})\right]^2$$

$$= \sigma^2/n + \mu^2$$

$$= \frac{n}{n-1}(\sigma^2 + \cancel{\mu^2}) - \frac{n}{n-1}\left(\frac{\sigma^2}{n} + \cancel{\mu^2}\right)$$

$$= \frac{n}{n-1}\sigma^2 - \frac{1}{n-1}\sigma^2 = \sigma^2$$

$$Var(Y) = E(Y^2) - \left[E(Y)\right]^2 \geq 0$$

$Var(Y) = 0 \rightarrow Y$ is constant with probability 1.

$E(Y^2) > \left[E(Y)\right]^2$ unless Y is constant

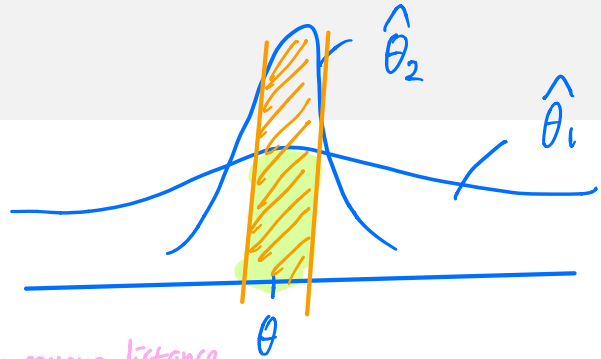
$\underbrace{E(S^2)}_{\sigma^2} > \left[E(S)\right]^2$ unless S is constant

In general, $\left[E(S)\right]^2 < \sigma^2$

$E(S) \neq 0$

MSE

measure goodness of pt estimator



★ Is Unbiasedness enough? X

Definition (Mean Square Error)

average square distance btw $\hat{\theta}$ & θ

$$MSE(\hat{\theta}) = E_\theta(\hat{\theta} - \theta)^2 = V(\hat{\theta}) + (B(\hat{\theta}))^2.$$

$\Rightarrow$ We want to use an estimator that has a small MSE.

Ex. $Y_1, \cdots, Y_n$: iid Unif(0, θ), $\Omega = \{\theta : 0 < \theta < \infty\}$

$$\hat{\theta}_1 = 2\bar{Y}, \hat{\theta}_2 = \frac{n+1}{n} \max Y_i$$

$\Rightarrow \hat{\theta}_2$ is better than $\hat{\theta}_1$ for all θ, if $n > 1$.

★ Assignment 2

$$E(\hat{\theta}-\theta)^2 = E(\hat{\theta} - \underline{E(\hat{\theta})} + \underline{E(\hat{\theta})} - \theta)^2$$

$$= E(\hat{\theta} - E(\hat{\theta}))^2 + 2E\left[(\hat{\theta} - E(\hat{\theta}))\underbrace{(E(\hat{\theta}) - \theta)}_{\text{constant}}\right]$$

$$+ E\left[(\underbrace{E(\hat{\theta})}_{\text{constant}} - \theta)^2\right]$$

constant! not random variable

$$= E(\hat{\theta} - E(\hat{\theta}))^2 + 2(E(\hat{\theta}) - \theta) \times \underbrace{E\left[\hat{\theta} - E(\hat{\theta})\right]}_{=0}$$

$$+ (E(\hat{\theta}) - \theta)^2$$

$E(Y - E(Y))$
$= E(Y) - E(Y) = 0$

$$= \underbrace{E(\hat{\theta} - E(\hat{\theta}))^2}_{Var(\hat{\theta})} + \underbrace{(E(\hat{\theta}) - \theta)^2}_{(\text{bias of } \hat{\theta})^2}$$

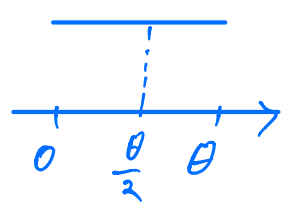
1. $Y_1, \ldots, Y_n$ iid unif$(0, \theta)$, $\theta > 0$

$$E(\bar{Y}) = E(Y) = \int_0^\theta \frac{y}{\theta}\, dy = \frac{y^2}{2\theta}\Big|_0^\theta = \frac{\theta}{2}$$

$$\downarrow$$

$$E(Y) = \int_{-\infty}^{\infty} y \underbrace{f_Y(y)}_{\frac{1}{\theta}} dy$$

pdf



$E(2\bar{Y}) = \theta$: unbiased estimator

W : $\max Y_i$

$$F_W(w) = P(\max X_i \le w)$$

$$= P(Y_1 \le w, \ldots, Y_n \le w)$$

$$= P(Y_1 \le w) \times \cdots \times P(Y_n \le w)$$

$$= [P(Y \le w)]^n = \begin{cases} 0 & \text{if } w \le 0 \\ \left(\frac{w}{\theta}\right)^n & 0 < w < \theta \\ 1 & \text{if } w > 0 \end{cases}$$

$$f_W(w) = \begin{cases} \dfrac{n w^{n-1}}{\theta^n}, & 0 < w < \theta \\ 0, & \text{otherwise} \end{cases}$$

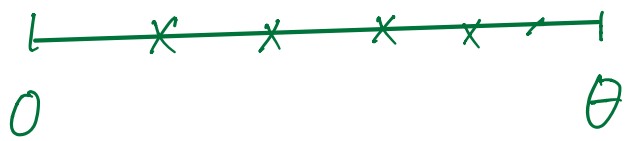
$$E(W) = \int_{-\infty}^{\infty} w \cdot f_W(w)\, dw$$

$$= \int_0^{\theta} w \cdot \frac{n w^{n-1}}{\theta^n}\, dw = \int_0^{\infty} \frac{n w^n}{\theta^n}\, dw$$

$$= \frac{n}{n+1} \frac{w^{n+1}}{\theta^n} \Big|_0^{\theta} = \frac{n}{n+1}\theta$$

not U.E because $E(W) \neq \theta$

bias of W $\quad E(W) - \theta = \frac{n}{n+1}\theta - \theta = -\frac{1}{n+1}\theta$



$$\underbrace{E\left(\frac{n+1}{n}W\right)}_{\text{unbiased!}} = \frac{n+1}{n}E(W) = \theta$$

$\hat{\theta}_1 = 2\bar{Y}$ $\qquad$ $\hat{\theta}_2 = \frac{n+1}{n}W$

여기서 $\hat{\theta}_1, \hat{\theta}_2$ 둘다 Bias = 0

$MSE = Var + (Bias)^2$

with UE, $MSE = Var$

$$Var(\hat{\theta}_1) = Var(2\bar{Y}) = 4\,Var(\bar{Y})$$

$$= 4 \times \frac{Var(Y)}{n} = \frac{4}{n} \times \frac{\theta^2}{12} = \frac{\theta^2}{3n}$$

$$U(a,b) \begin{cases} \text{mean} = \dfrac{a+b}{2} \\ \text{var} = \dfrac{(b-a)^2}{12} \end{cases}$$

$$Var(\hat{\theta}_2) = E(\hat{\theta}_2^{\,2}) - [E(\hat{\theta}_2)]^2$$

$$= E\left(\left(\frac{n+1}{n}\right)^2 W^2\right) - \theta^2$$

$$E(W^2) = \int_0^{\theta} w^2 \times \frac{n w^{n-1}}{\theta^n}\, dw$$

$$= \int_0^{\theta} \frac{n w^{n+1}}{\theta^n}\, dw = \frac{n}{n+2} \left.\frac{w^{n+2}}{\theta^n}\right|_0^{\theta}$$

$$= \frac{n\theta^2}{n+2}$$

$$= \frac{(n+1)^2}{n^2} \cdot \frac{n\theta^2}{n+2} - \theta^2$$

$$= \frac{(n+1)^2 - n^2 - 2n}{n(n+2)}\,\theta^2$$

$$= \frac{\theta^2}{n(n+2)}$$

$$MSE(\hat{\theta}_1) = \frac{\theta^2}{3n} \qquad MSE(\hat{\theta}_2) = \frac{\theta^2}{n(n+2)}$$

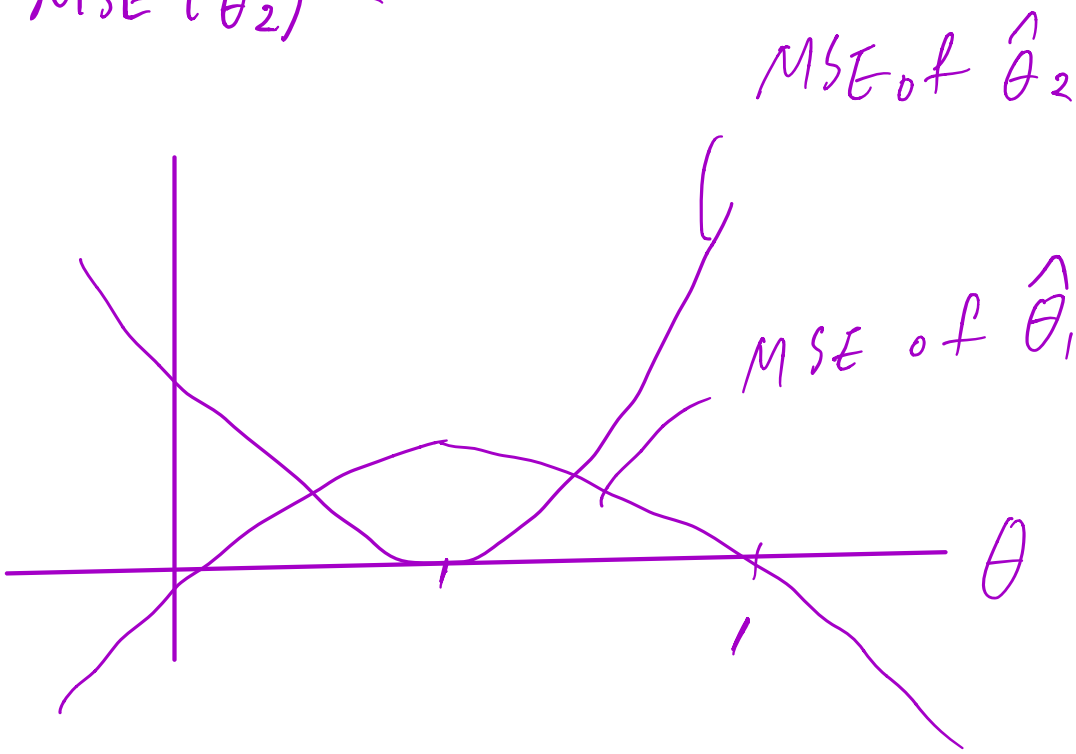
$$\frac{MSE(\hat{\theta}_1)}{MSE(\hat{\theta}_2)} = \frac{n+2}{3} > 1 \quad \text{when } n > 1$$

when $n > 1$, $MSE(\hat{\theta}_1) > MSE(\hat{\theta}_2)$

$n = 1$, $=$

$\boxed{\text{Exercise}}$

$X \sim \text{Bernoulli}(\theta)$ $\quad \hat{\theta}_1 = X$, $\hat{\theta}_2 = \frac{1}{2}$

$MSE(\hat{\theta}_1) =$

$MSE(\hat{\theta}_2) =$